

Allergic Rhinitis

What is Allergic Rhinitis?

We are currently in the middle of the spring allergy season. Remember the saying, “April showers bring May flowers”? The quote is attributed to a poem by Thomas Tusser in 1557. We had some early wet days in April, but overall, the month was not as wet as it has been in the past. This has moderated the allergens and pollen in the air to some extent.



Allergic rhinitis (AR) is an atopic disease characterized by symptoms of nasal congestion, clear rhinorrhea, sneezing, postnasal drip, and nasal pruritus. It affects one in six individuals and is associated with significant morbidity, loss of productivity, and healthcare costs. Historically, AR was thought to be a disease process of the nasal airway alone. Still, the development of the unified airway theory has classified AR as a component of systemic allergic response, with other associated conditions, such as asthma and atopic dermatitis, sharing an underlying systemic pathology.¹

AR can be classified as either seasonal (intermittent) or perennial (chronic), with approximately 20% of cases being seasonal, 40% perennial, and 40% with features of both.² In addition to nasal symptoms, patients with AR may also present with associated allergic conjunctivitis, non-productive cough, Eustachian tube dysfunction, and chronic sinusitis. Once diagnosed, AR is treatable with a variety of modalities, with intranasal glucocorticoids being the first-line therapy.¹

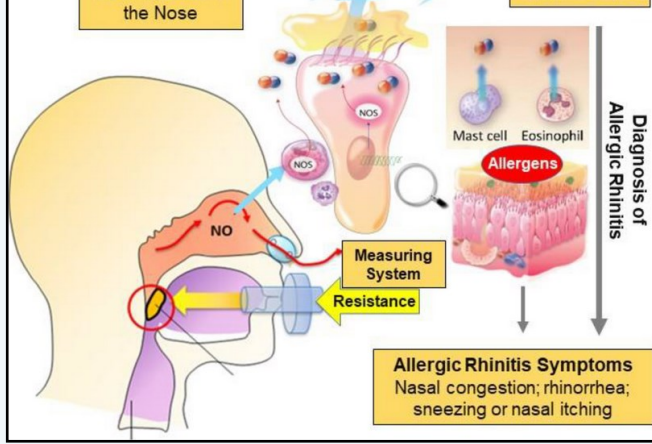
Etiology

The allergic response is classified into early and late-phase reactions. In the early phase, allergic rhinitis is an immunoglobulin (Ig)E-mediated response against inhaled allergens that cause inflammation driven by type 2 helper (Th2) cells.² The initial response occurs within five to 15 minutes of exposure to an antigen, resulting in the degranulation of host mast cells. This releases a variety of pre-formed and newly synthesized mediators, including histamine, which is one of the primary mediators of allergic rhinitis.

Histamine induces sneezing via the trigeminal nerve and also plays a role in rhinorrhea by stimulating mucous glands. Other immune mediators, such as leukotrienes and prostaglandins, are also implicated, as they act on blood vessels to cause nasal congestion. Four to six hours after the initial response, an influx of cytokines, such as interleukins (IL)-4 and IL-13, from mast cells occurs, signifying the development of the late-phase response. These cytokines, in turn, facilitate the infiltration of eosinophils, T-lymphocytes, and basophils into the nasal mucosa and produce nasal edema with resultant congestion.³

Non-IgE-mediated hyperresponsiveness can develop due to eosinophilic infiltration and nasal mucosal obliteration. The nasal mucosa now becomes hyperreactive to normal stimuli (such as tobacco smoke and cold air), and causes symptoms of sneezing, rhinorrhea, and nasal pruritus.⁴

There are data to suggest a genetic component to allergic rhinitis, but high-quality studies are generally lacking. Monozygotic twins show 45% to 60% concordance, and dizygotic twins have a concordance rate of approximately 25% in the development of AR. Specific regions on chromosomes 3 and 4 also correlate with allergic responses.⁵



Note: From Sinusitis [Infographic], by Sy Duong-Quy, 2022, MDPI (<https://www.mdpi.com/2673-351X/6/1/3>).

Epidemiology

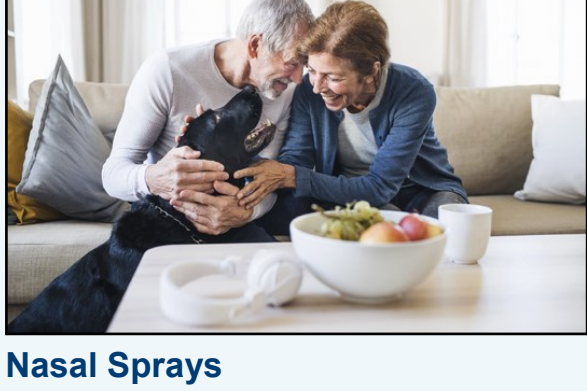
The prevalence of allergic rhinitis based on physician diagnosis is approximately 15%. However, based on patients with nasal symptoms, the prevalence is estimated to be as high as 30%. AR is known to peak in the second to fourth decades of life, then gradually decline.⁶

The incidence of AR in the pediatric population is also quite high, making it one of the most common chronic pediatric disorders. According to data from the International Study for Asthma and Allergies in Childhood, 14.6% in the 13- to 14-year age group and 8.5% in the 6- to 7-year age group display symptoms of rhinoconjunctivitis linked to allergic rhinitis.⁷ Seasonal allergic rhinitis seems to be more common in the pediatric age group, whereas chronic rhinitis is more prevalent in adults.⁸

Treatment and Management of Allergic Rhinitis

Avoiding Allergens

Avoidance of triggers, especially in those with seasonal symptoms, is encouraged, although it is not always practical. Precautions can be taken to avoid dust mites, animal dander, and upholstery, though



this can require significant lifestyle changes that may not be acceptable to the patient. If removing a pet from the home is not feasible, isolating the pet to a single room in the house may be an option to minimize dander exposure. It may take up to 20 weeks to eliminate cat dander from the home, even after removing the animal.

Allergen-impermeable bedding covers, washing sheets in hot water, and the use of a vacuum cleaner with high-efficiency particulate air (HEPA) filters may also lessen symptoms.⁵

Nasal Sprays

Pharmacological options include antihistamines, intranasal steroids, leukotriene receptor antagonists (LTRAs), and immunotherapy.

Intranasal corticosteroid therapy can be monotherapy or in combination with oral antihistamines in patients in patients with mild, moderate, or severe symptoms as early as 2 years of age. Studies have shown intranasal corticosteroids are superior to antihistamines in effectively reducing nasal inflammation and improving mucosal pathology. Thus, a topical intranasal steroid should be the first-line treatment for AR.⁹ Commonly available nasal sprays in the United States include beclomethasone, budesonide, fluticasone propionate, mometasone furoate, and triamcinolone acetonide.

Proper administration of nasal spray is critical in achieving an optimal clinical response and avoiding side effects; therefore, patients should always receive counsel on the appropriate use of devices. They should be used regularly, as their peak effect may take multiple days to develop. The spray bottle's tip should be placed just inside the nares and aimed laterally toward the ipsilateral eye to minimize contact of the product with the nasal septum. The most common side effect reported is nasal irritation, followed by epistaxis, both of which can be prevented by spraying away from the nasal septum.¹⁰ Oral and injectable steroids have been shown to alleviate symptoms of AR but are not recommended for routine use due to their significant systemic side-effect profile.¹¹



Antihistamines

First-generation antihistamines include diphenhydramine, chlorpheniramine, and hydroxyzine. Second-generation antihistamines include fexofenadine, loratadine, desloratadine, and cetirizine. Both first- and second-generation antihistamines are effective at controlling symptoms of AR. Still, first-generation antihistamines can be quite sedating due to their ability to cross the blood-brain barrier. These agents also act on muscarinic receptors, causing side effects of dry mouth, urinary retention, constipation, and/or tachycardia. **Pediatrics consideration:** Oral antihistamine should be reserved for children > 6 mos.

Second-generation antihistamines have improved H1 selectivity, are less sedating, and have longer half-lives (12 to 24 hours) than those of the first generation. Fexofenadine has no sedating effects, but loratadine and desloratadine may be sedating at higher doses. Cetirizine has the most potential for sedation of all second-generation antihistamines. There is no one agent recommended over others, as all have shown similar efficacy and safety profiles in terms of symptom relief.⁵

Intranasal antihistamines, such as azelastine, have a rapid onset and are more efficacious than oral antihistamines in relieving nasal symptoms. They are recommended as first or second-line therapies for AR and can be used in conjunction with topical nasal steroid sprays with a synergistic effect.¹¹

Other Treatment Options

Leukotriene receptor antagonists (LTRAs), such as montelukast and zafirlukast, can be beneficial in patients with AR, but they are not as efficacious as intranasal corticosteroids.¹² As such, their use should be limited in virtual care.



Oral decongestants, such as pseudoephedrine, are useful in relieving symptoms but are not recommended for extended daily use due to their side-effect profile. Intranasal decongestants, such as xylometazoline, are alpha agonists that are delivered directly to nasal tissue to produce vasoconstriction. Prolonged use of intranasal decongestants has a risk of causing rebound nasal congestion (rhinitis medicamentosa) and, therefore, should not be used for more than a week.¹¹

Sodium cromoglycate (Cromolyn) effectively reduces sneezing, rhinorrhea, and nasal pruritus, so it is a reasonable option. Surgical treatment is reserved for patients with nasal polyposis, inferior turbinate hypertrophy causing intractable nasal obstruction, or chronic sinus disease refractory to medical treatment.⁵ Budesonide is the only FDA-approved agent for pregnant patients experiencing symptoms of allergic rhinitis.¹

Omalizumab, a monoclonal antibody, is beneficial in patients with AR, although the cost associated with therapy is a limiting factor in its use.¹³ Nasal saline can be another option in conjunction with other treatment modalities. Isotonic solutions are more beneficial in adults, whereas hypertonic solutions may be more effective in children.¹¹

Where Can I Learn More About Allergic Rhinitis?

For much more information, please refer to the [Clinical Practice Guidelines: Allergic Rhinitis](#) published by the Otolaryngology-Head and Neck Surgery Society.

Appendix

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